

Customer Churn Dataset — Analytics Report

[Business Analytics Report](#)

Project: Customer Churn Dataset

Dataset: customer_churn_dataset (1).csv

Analysis date: 2026-09-25

Prepared by: kabeer ali

Generated by the AI Data Scientist Platform

Table of Contents

1. Executive Summary	4
Key verified findings	4
Top recommendations	4
2. About the Dataset	5
Column dictionary	5
3. Data Quality & Cleaning	7
Original dataset quality (as uploaded)	7
Duplicate Rows	7
Invalid Dates	7
Missing Values	7
Outliers	7
Impossible Values	8
Cleaning performed	8
4. Key EDA Findings	10
Trend over time (Signup Date)	10
Support Tickets breakdown	11
5. Modeling Approach	12
Features used	12
Features excluded, and why	12
6. Model Performance	13
Model comparison	13
In plain language	13
Confusion Matrix	13
7. What Drives the Outcome	15
8. Future Outlook	16
9. AI Business Insights AI-generated	17
10. Limitations	18
11. Appendix	19
Full model metrics	19
Preprocessing	19
Hyperparameters	19
Dataset statistics	19

1. Executive Summary

This report analyzes **customer_churn_dataset (1).csv** (2,000 rows, 17 columns) covering 2020-07-02 to 2026-05-31 to answer the business question: *A synthetic dataset of 2,025 subscription customers with 17 columns covering demographics, subscription plan, contract type, payment method, tenure, spending, purchase activity, support tickets, and satisfaction scores. The target column "churn" shows whether the customer left (1) or stayed (0). It includes some missing values, duplicate rows, outliers, and invalid dates for testing data cleaning. Suitable for churn classification and revenue regression..*

Key verified findings

- The original dataset contained 344 missing value(s) and 25 duplicate row(s), which have since been cleaned.
- A XGBoost model was trained to predict **Churn** (classification). The model correctly classifies about 71.0% of cases (Accuracy).

Model reliability, in plain terms: The model correctly classifies about 71.0% of cases (Accuracy). F1 score (balancing precision and recall) is 0.697. ROC-AUC is 0.757 (1.0 = perfect separation, 0.5 = no better than chance).

Top recommendations

- The most common observed outcome for Churn is '0' (259 of 400 test cases).
- Among customers, the 'Satisfaction Score: Lower' group shows the highest observed rate (43.9%, n=667), compared to 'Satisfaction Score: Higher' (29.2%, n=667).

2. About the Dataset

customer_churn_dataset (1).csv contains 2,000 rows and 17 columns. It spans the period 2020-07-02 to 2026-05-31 (based on Signup Date).

Business question: A synthetic dataset of 2,025 subscription customers with 17 columns covering demographics, subscription plan, contract type, payment method, tenure, spending, purchase activity, support tickets, and satisfaction scores. The target column "churn" shows whether the customer left (1) or stayed (0). It includes some missing values, duplicate rows, outliers, and invalid dates for testing data cleaning. Suitable for churn classification and revenue regression.

Note: the project description mentions Target, but no corresponding column was found in this dataset — this report cannot speak to that directly.

Column dictionary

Column	Type	Missing	Notes
Customer Id	Identifier	0.0%	
Signup Date	Date/Time	0.0%	
Age	Numeric	0.0%	typical range 18.00–70.00
Gender	Category	0.0%	most common: Male
Region	Category	0.0%	most common: North
Subscription Plan	Category	0.0%	most common: Standard
Contract Type	Category	0.0%	most common: Month-to-Month
Payment Method	Category	0.0%	most common: Bank Transfer
Tenure Months	Numeric	0.0%	typical range 1.00–72.00
Monthly Spend	Numeric	0.0%	typical range 2.12–482.80
Purchase Frequency	Numeric	0.0%	typical range 0.00–19.00

Days Since Last Purchase	Numeric	0.0%	typical range 0.00–208.00
Support Tickets	Numeric	0.0%	typical range 0.00–7.00
Satisfaction Score	Numeric	0.0%	typical range 1.00–5.00
Discount Used	Category	0.0%	most common: No
Total Revenue	Numeric	0.1%	typical range 15.18–6,662.3
Churn	Numeric	0.0%	typical range 0.00–1.00

3. Data Quality & Cleaning

Original dataset quality (as uploaded)

Total rows	Missing cells	Duplicate rows	Issues detected
2,025	344	25	16

Duplicate Rows

Column	Finding	Affected
(whole dataset)	25 duplicate row(s) detected.	25

Invalid Dates

Column	Finding	Affected
signup_date	'signup_date' has 4 value(s) that don't parse as a valid date.	4

Missing Values

Column	Finding	Affected
age	'age' has 80 missing value(s) (4.0%).	80
region	'region' has 40 missing value(s) (2.0%).	40
payment_method	'payment_method' has 60 missing value(s) (3.0%).	60
monthly_spend	'monthly_spend' has 61 missing value(s) (3.0%).	61
satisfaction_score	'satisfaction_score' has 103 missing value(s) (5.1%).	103

Outliers

Column	Finding	Affected
tenure_months	'tenure_months' has 79 statistical outlier(s) outside [-19.0, 61.0].	79
monthly_spend	'monthly_spend' has 15 statistical outlier(s) outside [-20.14, 106.12].	15
purchase_frequency	'purchase_frequency' has 4 statistical outlier(s) outside [-2.0, 14.0].	4

days_since_last_purchase	'days_since_last_purchase' has 95 statistical outlier(s) outside [-44.5, 95.5].	95
support_tickets	'support_tickets' has 132 statistical outlier(s) outside [-0.5, 3.5].	132
satisfaction_score	'satisfaction_score' has 22 statistical outlier(s) outside [1.5, 5.5].	22
total_revenue	'total_revenue' has 131 statistical outlier(s) outside [-1207.06, 2993.9].	131

Impossible Values

Column	Finding	Affected
monthly_spend	'monthly_spend' has 2 negative value(s) that are rare exceptions among otherwise non-negative values — likely a data-entry error, not real signal.	2
total_revenue	'total_revenue' has 2 negative value(s) that are rare exceptions among otherwise non-negative values — likely a data-entry error, not real signal.	2

Cleaning performed

Problem	Before	Action	After	Reason
Duplicate rows	25 duplicate row(s)	Removed duplicate rows	0 duplicate rows	Duplicate records would otherwise double-count the same transaction/customer.
Missing Age	80 missing	Filled using the column median	0 missing	Reduces the influence of extreme values compared to the mean.
Missing Region	40 missing	Filled using the most frequent value (mode)	0 missing	The most representative value for a categorical field.
Missing Payment Method	60 missing	Filled using the most frequent value (mode)	0 missing	The most representative value for a categorical field.
Missing Monthly Spend	61 missing	Filled using the column median	0 missing	Reduces the influence of extreme values compared to the mean.
Missing Satisfaction Score	103 missing	Filled using the column median	0 missing	Reduces the influence of extreme values compared to the mean.

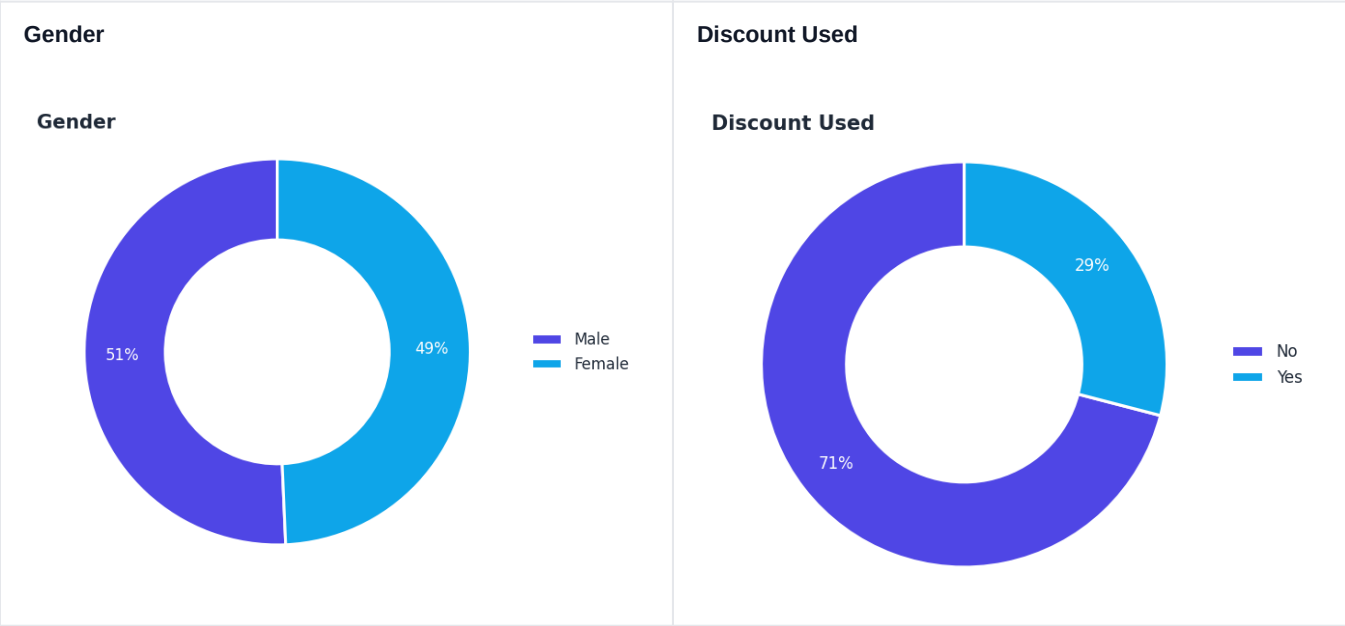
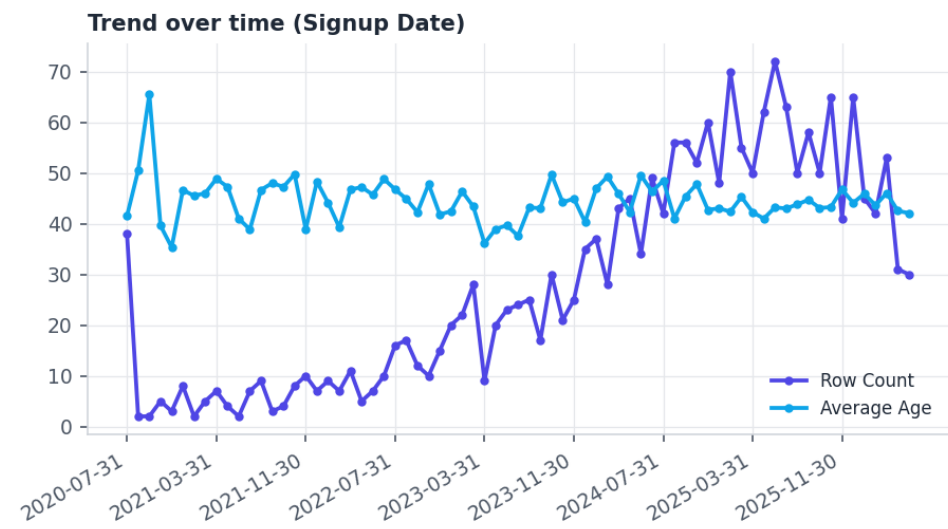
Impossible negative Monthly Spend	2 impossible value(s)	Converted to missing, then filled	0 impossible values	A rare negative value in an otherwise non-negative column is almost always a data-entry error.
Impossible negative Total Revenue	2 impossible value(s)	Converted to missing, then filled	0 impossible values	A rare negative value in an otherwise non-negative column is almost always a data-entry error.

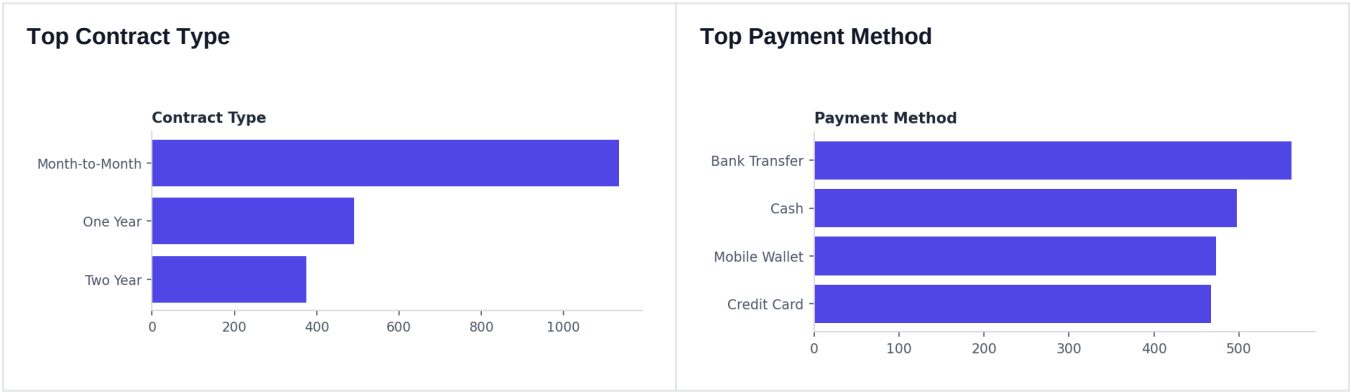
After cleaning: 2 missing cell(s), 0 duplicate row(s), 2,000 rows remain. The original file itself was never modified — cleaning produced a separate working copy.

4. Key EDA Findings

Total Rows	Churn Rate	Total Age	Total Tenure Months
2,000	35.2%	88,143	46,396

Trend over time (Signup Date)





Support Tickets breakdown

Average: 1.47 (range 0.0–7.0)

Value	Count	Percent
7	1	0.1%
6	5	0.2%
5	34	1.7%
4	89	4.5%
3	226	11.3%
2	501	25.1%
1	691	34.5%
0	453	22.7%

5. Modeling Approach

A **classification** model was trained to predict **Churn**, using 17 feature(s).

Features used

Signup Date: Year, Signup Date: Month, Signup Date: Dayofweek, Age, Gender, Region, Subscription Plan, Contract Type, Payment Method, Tenure Months, Monthly Spend, Purchase Frequency, Days Since Last Purchase, Support Tickets, Satisfaction Score, Discount Used, Total Revenue

Features excluded, and why

Column	Reason excluded
Customer Id	Identifier column (e.g. an ID) — carries no predictive signal.
Signup Date	Raw date — replaced with derived year / month / day-of-week features instead.

Re-included at the user's request: Satisfaction Score

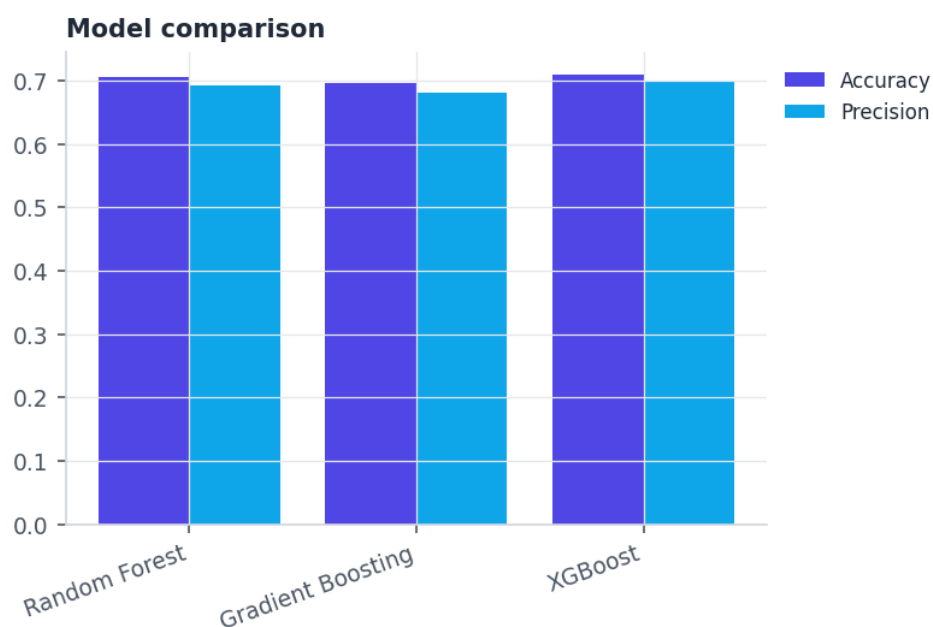
Technical details

Algorithm: XGBoost · Preprocessing: 11 numeric + 6 categorical feature(s), scaled/encoded · Train/test split: 20.0% held out · Cross-validation folds: 5 · Random seed: 42

6. Model Performance

Model comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Random Forest	0.7050	0.6923	0.7050	0.6873	0.7463
Gradient Boosting	0.6950	0.6802	0.6950	0.6721	0.7556
XGBoost	0.7100	0.6987	0.7100	0.6973	0.7566

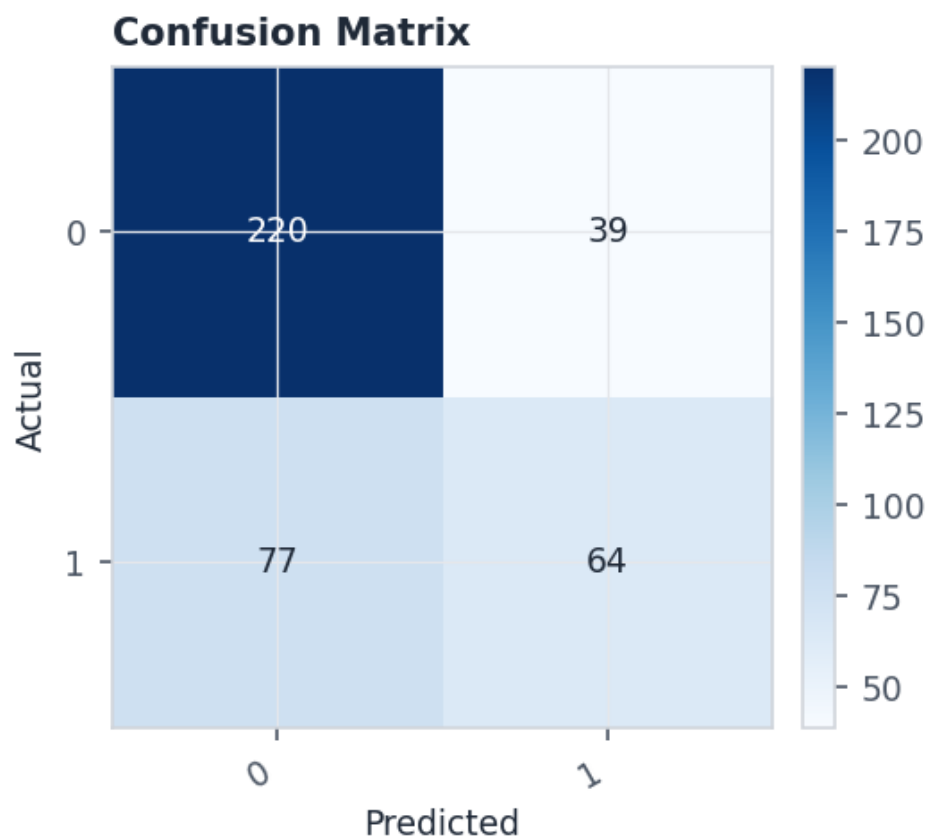


In plain language

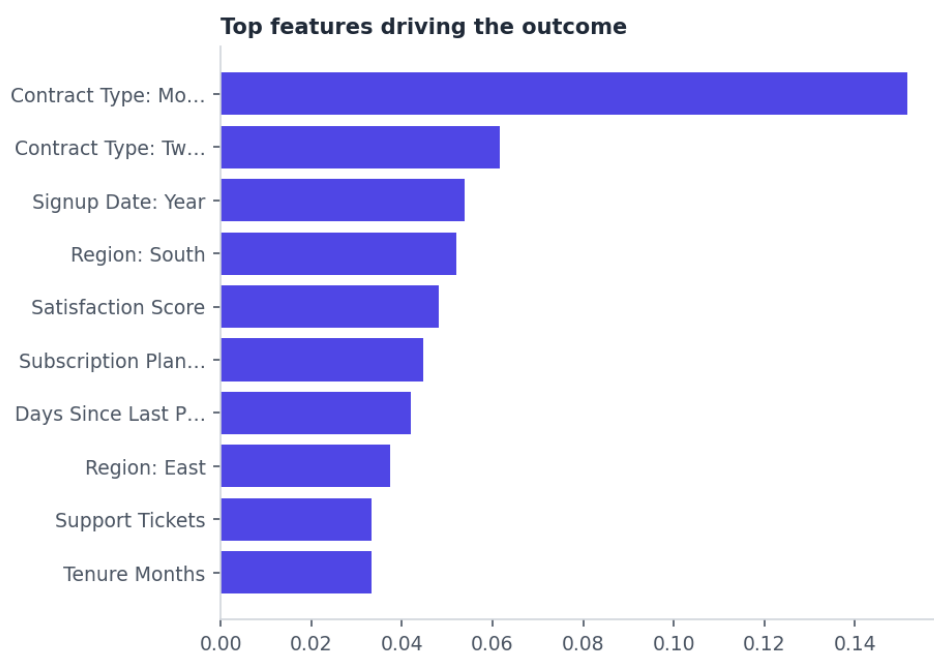
- The model correctly classifies about 71.0% of cases (Accuracy).
- F1 score (balancing precision and recall) is 0.697.
- ROC-AUC is 0.757 (1.0 = perfect separation, 0.5 = no better than chance).

Compared against a trivial baseline (F1 Score = 0.509): this model clearly outperforms it.

Confusion Matrix



7. What Drives the Outcome



Feature	Relative importance
Contract Type: Month to Month	15.2%
Contract Type: Two Year	6.2%
Signup Date: Year	5.4%
Region: South	5.2%
Satisfaction Score	4.8%
Subscription Plan: Premium	4.5%
Days Since Last Purchase	4.2%
Region: East	3.7%
Support Tickets	3.3%
Tenure Months	3.3%

8. Future Outlook

- The most common observed outcome for Churn is '0' (259 of 400 test cases).
- Among customers, the 'Satisfaction Score: Lower' group shows the highest observed rate (43.9%, n=667), compared to 'Satisfaction Score: Higher' (29.2%, n=667).

Group	Observed rate	Group size
Satisfaction Score: Lower	43.9%	667
Satisfaction Score: Middle	32.6%	666
Satisfaction Score: Higher	29.2%	667

9. AI Business Insights AI-generated

AI insights are unavailable right now (gemini: Gemini rate limit exceeded — too many requests. Try again shortly.; ollama fallback: The AI service is currently unavailable.). The rest of this report contains verified analytical results only.

10. Limitations

- Cross-validation scores varied by ± 0.010 across folds — treat the headline metric as an estimate with this much natural uncertainty, not an exact figure.
- Model predictions are estimates based on historical patterns and may not hold if underlying business conditions change. They should inform, not replace, domain expertise.

11. Appendix

Full model metrics

Metric	Value
Accuracy	0.71
Precision	0.6987
Recall	0.71
F1 Score	0.6973
ROC-AUC	0.7566
Cv Mean	0.7073
Is Weak	False

Preprocessing

Setting	Value
Test Size	0.2
Cv Folds	5
Random seed	42
Model version	6

Hyperparameters

Parameter	Value
Random Seed	42

Dataset statistics

Statistic	Value
Rows	2,000
Columns	17